A Model-Driven Approach For Information System Migration

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Outline

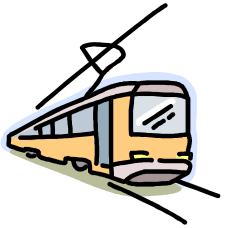
- The TRAMs project (2002-2003)
- RM-ODP Standard as an Intermediate Metamodel (IMM)
- The COBOL to JAVA Demonstrator
- Coupling RM-ODP and Action Semantics standards
- Conclusion and further works

Aims

- Define a methodology
- Provide an open and modular migration framework

In order to

- Insure IS migrations in a cost effective manner
- Experiment metamodels and model transformation techniques



EDF Electricité de France

france telecom

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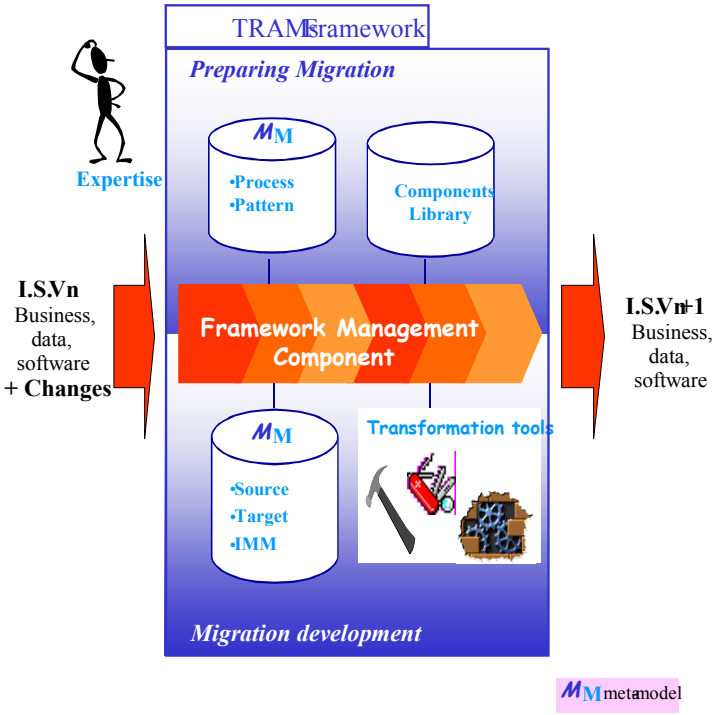
SOFT-MAINT GROUPE SODIFRANCE

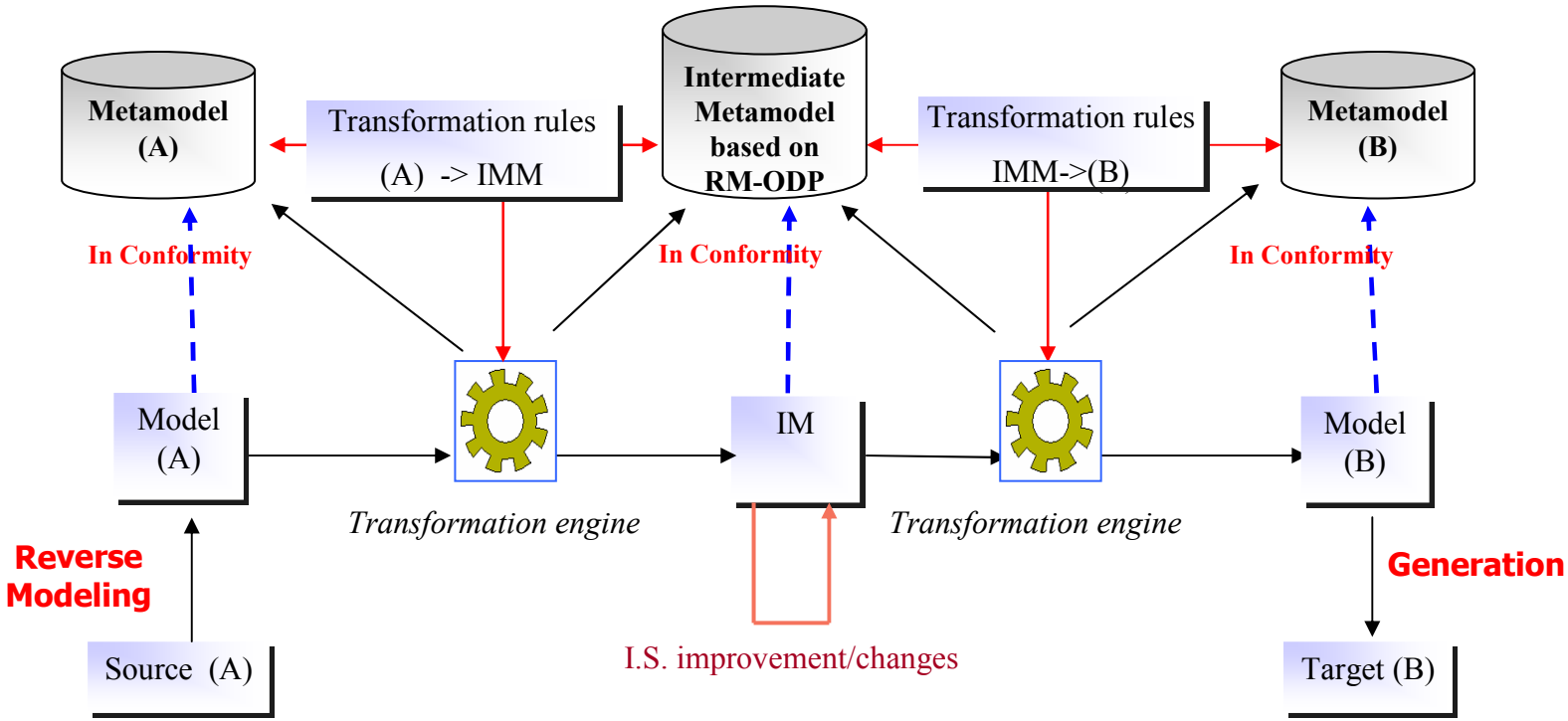
RNTL logiciel

TRAMs: a methodology and a Framework

- The *preparing migration* phase
  - Identify the migration nature (business or technologic)
  - Identify the Metamodels to be used
  - Model the migration process (eg. in SPEM)
  - Specify the transformation components
- The *execution* phase
  - Monitored by a management component
  - Insured by several tools and components
  - Includes the execution of manual actions/interactions



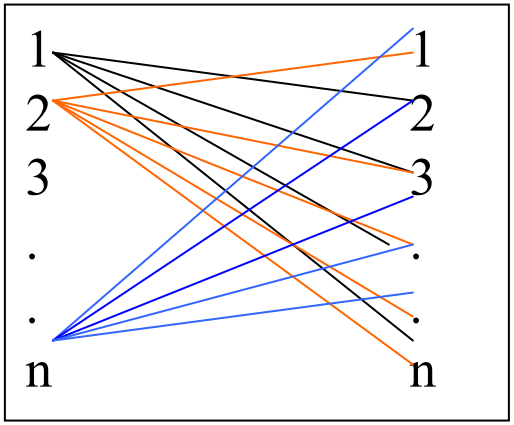
The TRAMs generic migration-process



Why an Intermediate Metamodel?

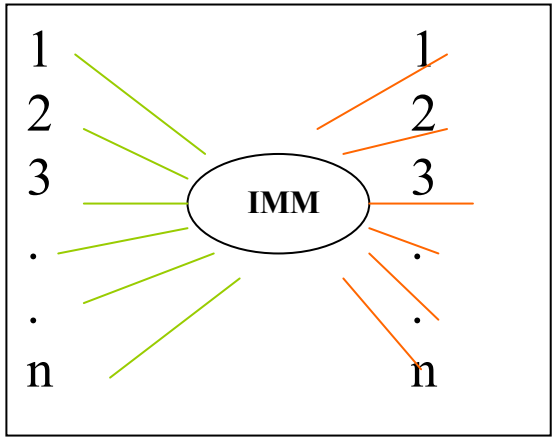
- To decrease significantly the number of model transformers to be developed
- To decrease time and cost for further migrations of the Information System

Point to point Migration



For n formalisms → n(n-1) set of transformation rules

Migration using an Intermediate Metamodel



For n formalisms → Only 2n set of transformation rules

IMM Specification : RM-ODP standard

- RM-ODP (Reference Model Open Distributed Processing) as an Intermediate Metamodel
- Why RM-ODP?

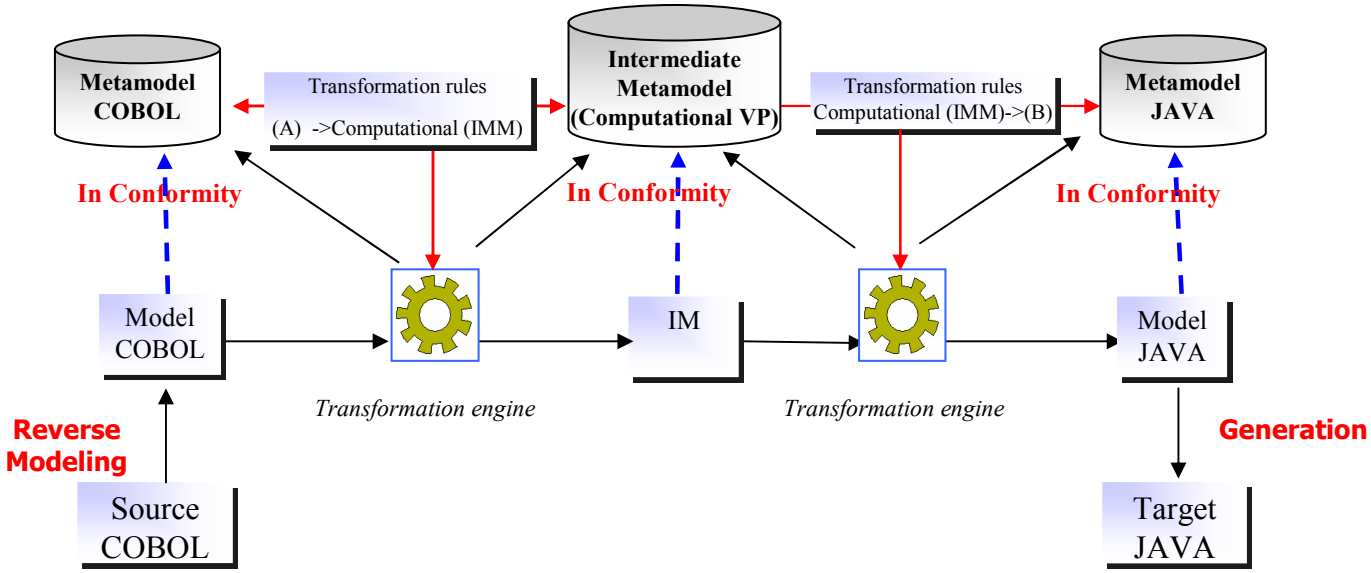
  - Generic, stable and rigorous concepts

  - Concern separation based System Specification: the *Viewpoints* (Enterprise, Information, Computational, Engineering et Technology)

  - Used in several domains.

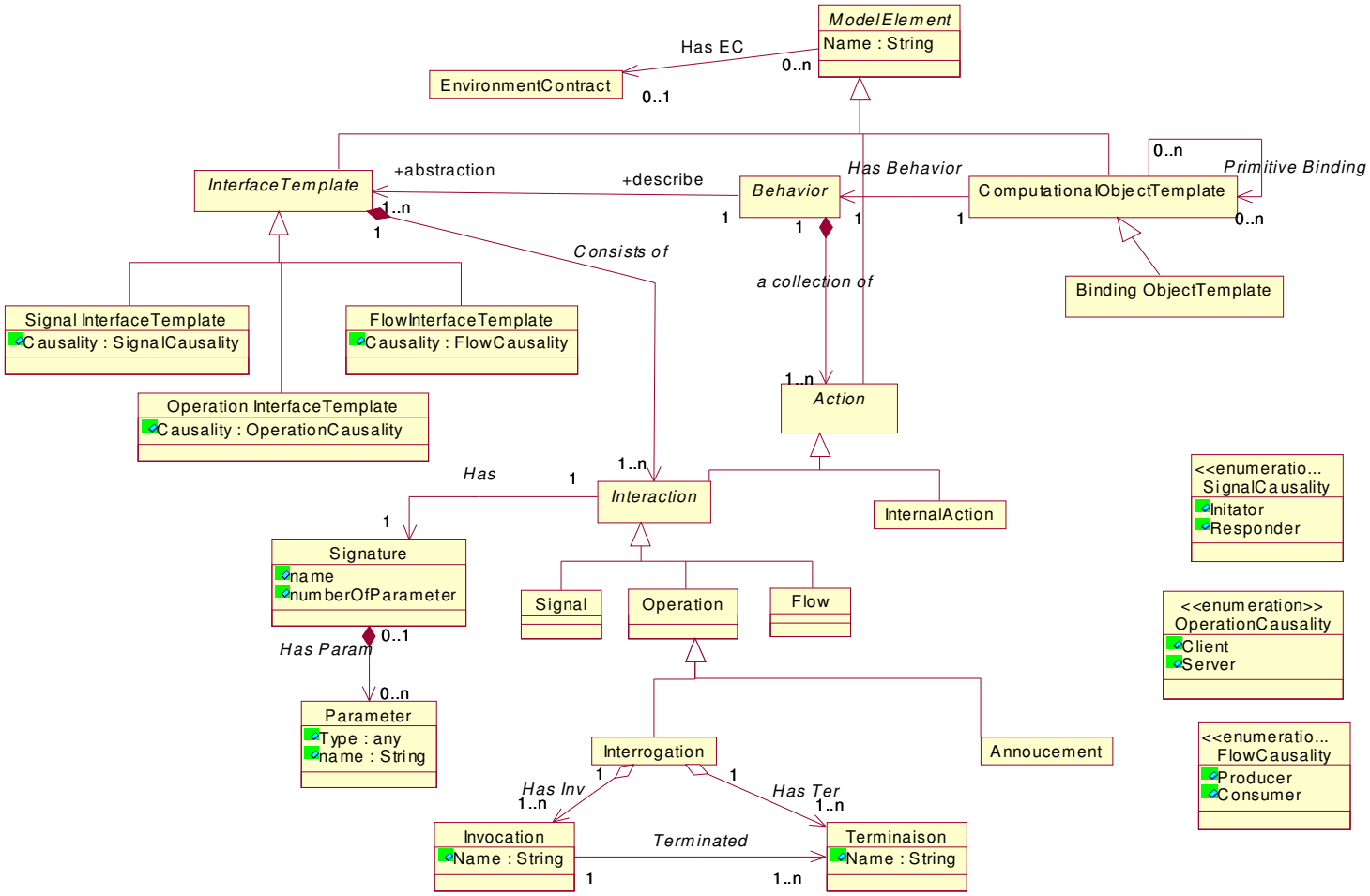
The COBOL to Java Demonstrator

- Migration of a sample of an insurance company application
- Focusing on the user interface with limited business features
- Choice of RM-ODP *Computational* Viewpoint as an Intermediate Model



COBOL to Java migration using RM-ODP Computational Viewpoint

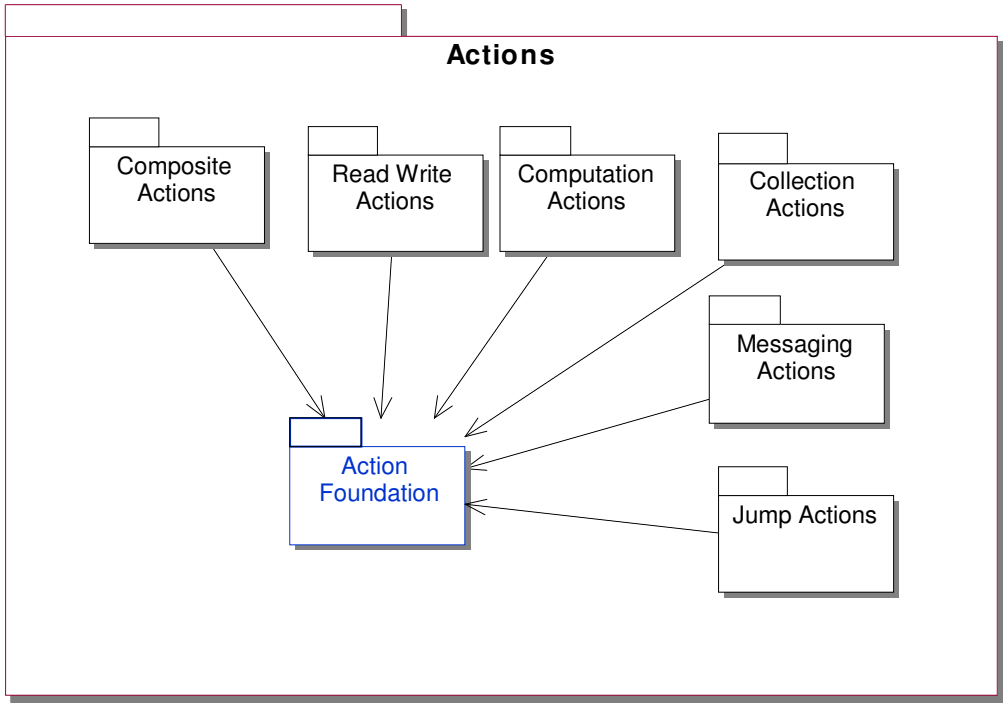
The ODP Computational viewpoint



The Computational Viewpoint Metamodel

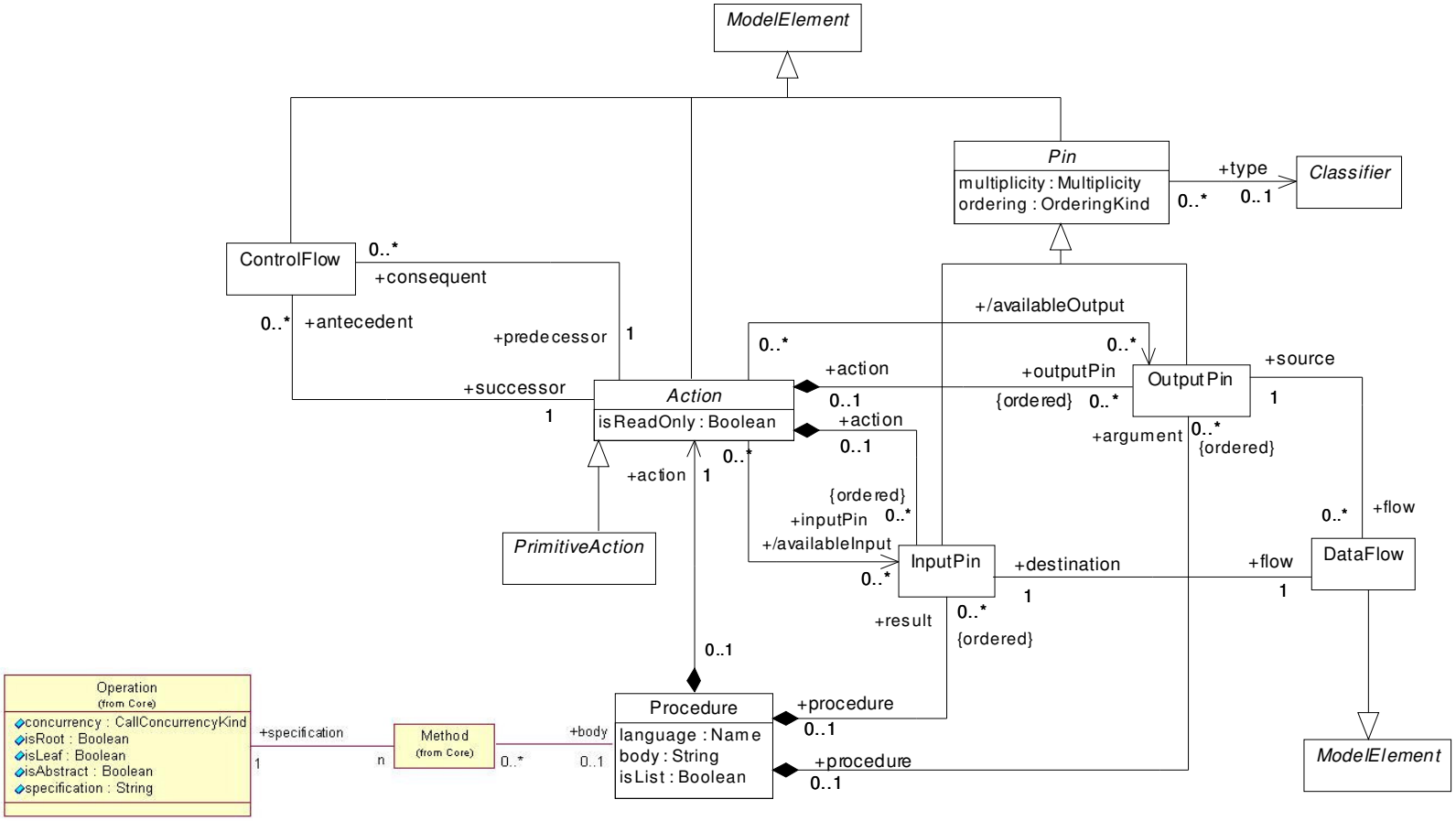
UML Action Semantics standard

- Action Semantics: an extension to UML
- Allows a precise specification of the behavior
- Comes in form of a package: ***Actions***

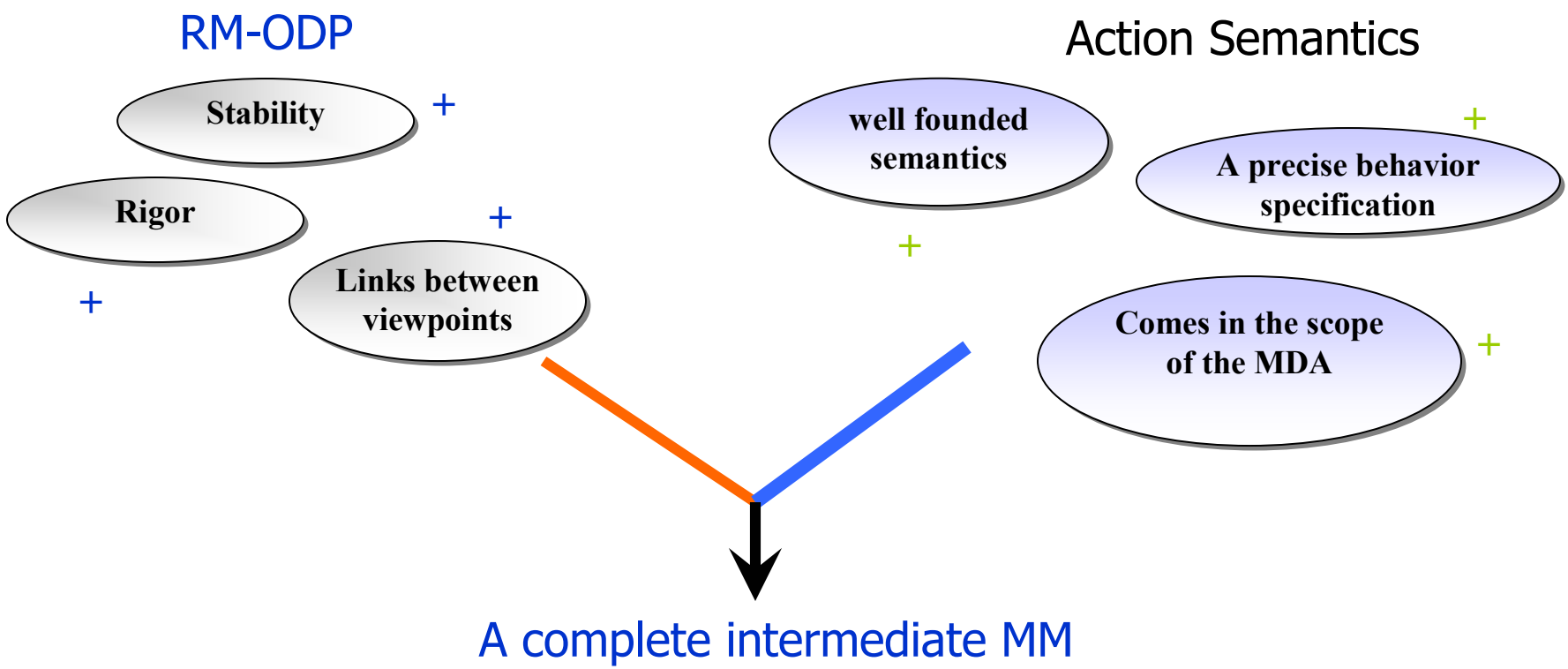


The Action Foundation package

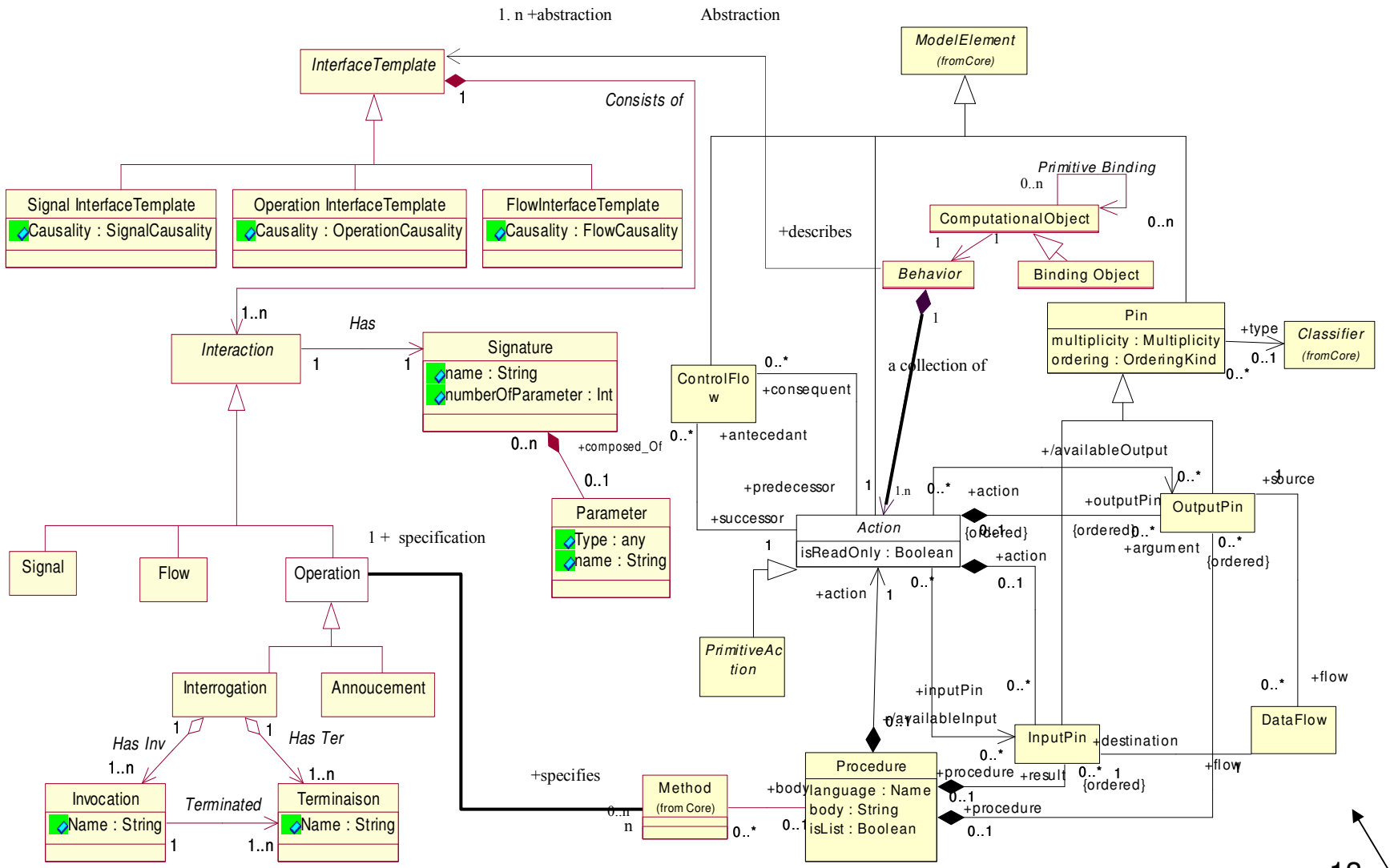
- The *Action Foundation* package: the core of Action Semantics



RM-ODP vs Action Semantics

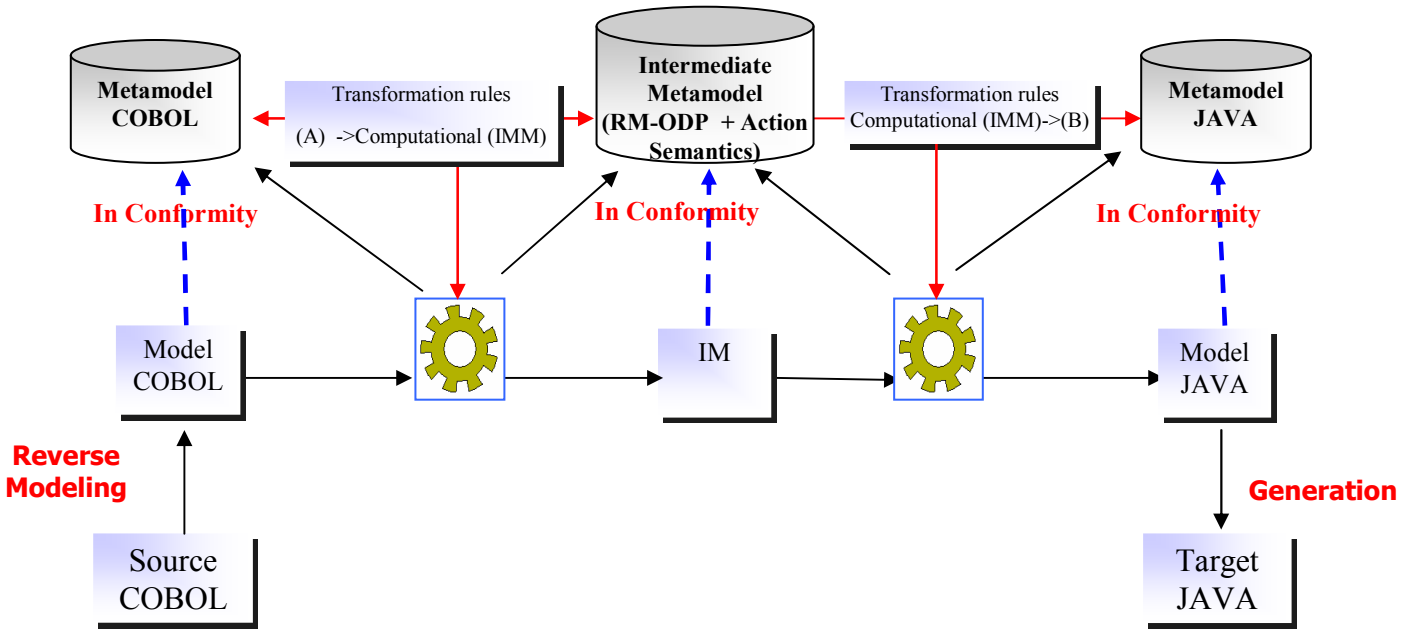


Integrating the two standards



The final COBOL to Java Demonstrator

- An optimized migration with precise rules



COBOL to Java migration using RM-ODP and Action Semantics

Conclusion and further works

- TRAMs framework and methodology have
  - Improved existing migration tools and best practices
  - Influenced OpenQVT
- TRAMs Intermediate MM
  - Concrete semantics not needed
  - Good complementarity of the two standards
- Still to come…
  - Improve code analysis
  - Extend the business aspects in the migration